Assignment # 2

STAT 679: Statistical Methods for Spatial Lattice Data–Spring 2026, UW-Madison

1. **(10 pts)** Refer to *Phosphorous Levels Data Example, Phosphorous Example B: Exploratory Data Analysis*. Use Geary's c to test for spatial correlation in both organic and inorganic phosphorous for Phosphorous Example B under randomization. Interpret your findings.

2. **(10 pts)** Refer to *Phosphorous Levels Data Example, Phosphorous Example B: Standard Linear Regression*. Using Phosphorous Example B, perform the residual analysis for the standard linear regression using different weights for your list of spatial weights. Comment on the similarities and differences in the model output. Discuss a case where you would want to use your proposed alternative weight.

3. **(15 pts)** The data set `wheat.csv` (on Canvas) lists yields of grain and straw from 500 plots of wheat laid out in a 25×20 lattice covering approximately 1 acre in total area. The data were collected by Mercer and Hall in 1910 at the Rothamsted Experimental Station in England.

   (a) Make a contour plot of the grain measurements. A quick way to do this is as follows:

   ```
   wheat <- read.csv("wheat.csv")
   z <- matrix(wheat$grain, nrow = 25, byrow = TRUE)
   contour(1:25, 1:20, z)
   ```

   (b) Perform exploratory data analysis on this dataset. Include summary statistics and at least one visualization. Provide 1–2 sentences on what you find.

   (c) Calculate Moran's I for grain and perform testing for

   $$H_0 : \text{CSR} \quad \text{vs.} \quad H_A : \text{Not CSR}$$

   based on a "rook's case" and on a "queen's case." Interpret your results. Are they consistent with your findings in part (a)?